

Augmenting a Process Model with State Machine and Business Artifacts*

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Abstract—Business processes play a central role in today's smart applications. To become even smarter, they should be flexible, executable and measurable for adapting to changing demand, being enacted effectively and assessed precisely. Due to the fact that de-facto process modeling languages do not cover these properties, this limitation gives rise to substantial amount of work for the extensions of the mainstream process modeling languages. The existing extensions support to capture the other perspectives of processes (e.g., resource allocation, performance measurement). Focusing on data flow aspect, we propose a framework for augmenting a control flow process model to yield an artifact-centric process model based on the state machine and business artifacts. The proposed mechanism is presented with the help of a real-life case study provided by our industrial partners.

Index Terms—State Machine, Artifact-Centric Process, Process Modeling.

1. Introduction

In the era of Internet of Things and Industry 4.0, the trend is to make everything smarter, e.g., smart factories, smart cities, smart universities, smart houses, smart hospitals. Business processes play a central role in today's smart applications. A process should be flexible, executable and measurable for the following reasons. First, business processes are subject to change, i.e., adapting its internal resources to fit the external requirements [1], [2], [3]. Second, business processes are increasingly enacted in an automatic way [4]. Third, the effectiveness of a process should be quantified for measuring its performance, e.g., time and cost [5] [6]. In fact, the existing visualization methods in process modeling like BPMN (Business Process Model and Notation), UML (Unified Modeling Language) activity diagram, etc. could not satisfy the above issues. Therefore, the business process models need to be enriched.

BPMN is one of the most notable languages in process modeling because it offers a strong visualization notation system and it is popular among domain experts. It leads to extending BPMN receiving a lot of attention from the

research communities. For modeling resource perspective requirements, the extensions of BPMN support to show how resource allocation can be captured [7] [8] and express constraints of resource by using the Object Constraint Language (OCL) [9]. In the performance measurement aspect, BPMN meta-model could be extended to depict the performance analysis (e.g., time, cost) of business processes [5] [6]. One of the most notable extensions of BPMN is used to express data flow in business processes [10] that only focused on describing the input and output objects but lacked additional information of states and attributes of the such artifacts. It is the motivation for us to investigate how business processes capture the states and attributes of the said artifacts.

For augmenting business processes with business artifacts, ACP (Artifact-Centric Process) models were launched to present the business process in an artifact-centric view. Estañol et al. proposed that ACP model was defined in terms of the Balsa framework including four "dimensions": Business Artifacts, Lifecycles, Services and Associations, which are mapped into UML diagrams [11]. However, it still requires a lot of manual activities. To solve the such issue, we would like to propose a formal framework for generating pre-conditions and post-conditions in ACP models (semi-)automatically.

We apply our framework to a real-life company, is a low-cost domestic airfare retailer in Vietnam, for testing its feasibility and accuracy. The aforementioned company wish to enhance their performance by redesigning their business processes. Therefore, our framework aims to generate the ACP models as the inputs of the redesign framework in question.

The remaining of this paper is structured as follows. We give our research background in Section 2. Section 3 expresses our research motivation by analyzing a running example and formulating our research questions. Furthermore, Section 4 is dedicated to the main contribution of our work including general blocks and a formal framework. We survey related work in Section 5. Section 6 concludes the paper and outlines our further work.

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2. Background

2.1. Artifact-Centric Process Model

Business process modeling traditionally put a lot of effort into the behavioral aspect. The so-called activity-centric modeling community became very crowded, resulting in several initiatives for both imperative and declarative modeling of the control-flow of business processes. Making an exhaustive list of them is not the goal of this paper. To name just a few, we have BPEL¹ (Business Process Execution Language), YAWL² (Yet Another Workflow Language), UML³ activity diagram, Petri net, etc.

Artifact-centric process (ACP) model takes a different approach to modeling business processes [12] [13]. Conceptually, both business objects and the process control are taken into account in models, resulting in an explicit representation of the transformation of business artifact in contract-like description⁴ of business tasks [15]. This approach has been further researched in connection to the UML and the Business Process Model and Notation (BPMN)⁵.

2.2. Extended BPMN

In business process aspect, there exists varied modeling languages, e.g., UML activity diagram, flowchart, BPMN. The most notable is BPMN that provides a standard and adequate notation system for modeling business processes, so it is popular among domain experts. The current BPMN standard provides two divergent aspects on business processes: collaboration diagrams, choreography diagrams [10].

However, in some specific situations, the standard BPMN fails to support; therefore it needs to be extended. For example, to capture advanced choreography scenarios, Decker and Puhmann [16] proposed expansions to widen the applicability of BPMN using the Services Interaction Patterns. Whereas, Momotko and Nowicki [17] wish to expand the BPMN by focusing on the visualization of process executions based on showing the process history, its current state and feasible future execution. For modeling artifact-centric processes, Lohmann and Nyolt [10] proposed an approach to expand BPMN is suitable for capturing data flow in business processes. So we base our work on this approach to put more information into the business processes for connecting it with other diagrams in our model transformation.

1. BPEL Specification <https://www.oasis-open.org/committees/wsbpel>

2. YAWL Foundation <http://www.yawlfoundation.org/>

3. UML Specification <http://www.omg.org/spec/UML/>

4. Each business task is described in terms of pre-conditions (i.e., conditions that must hold for the task to be invoked) and post-conditions (i.e., conditions that will be held upon the completion of the task) [14].

5. BPMN Specification <http://www.bpmn.org/>

3. Research Motivation

3.1. Running Example

We describe in this subsection a real case-study of a medium-sized company, nicknamed AD, that retails low-cost domestic airfare in Vietnam including Vietnam Airlines⁶, Jetstar Pacific Airlines⁷ and VietJet Air⁸. The company's vision is to keep improving its operational levels by means of improving employee competencies, optimizing business processes and technological innovations. The company has approximately 30 employees divided into four departments, namely finance, customer support, technical, marketing. The company deals with domestic airlines to allow its customers to book air tickets primarily over the Internet. It offers customer support services to those who wish to revise their air tickets at their own expenses.

In this case-study, we take a close look at the operationalization of this business capability dealing with the most frequently requested changes made to air tickets: flight rescheduling.

Figure 1 gives a detailed representation of the business process of handling rescheduling requests in the Business Process Modeling Notation (BPMN)⁹. The customer, company AD and airlines providing airfare (under role *Provider*) are each put under a BPMN pool in this diagram. The customer first makes a request to reschedule her air ticket. Company AD (actually its department of customer support) then checks the availability for her newly-proposed flight reschedule and gets back to her with reschedule fees incurred if available. Otherwise, company AD makes a list of reschedule options checking associated fees with her again. This activity is to be repeated until the customer agrees on her rescheduled flight and the associated fees. Next, company AD asks her to pay a surcharge, usually higher¹⁰ than the confirmed reschedule fees. Company AD then proceeds with payment for its provider and determines if they need to refund the customer. She receives a notification about the airfare reschedule (and possibly a refund) at the end of the process.

The board of managers of this company wish to improve their performance by changing their business processes. However, their processes currently are described in visualization forms that leads to lack of information to ensure the accuracy of the such models for (semi-)automatic re-engineering frameworks. That is the motivation for us to propose a framework for transforming the method of

6. <http://www.vietnamairlines.com>

7. <http://www.jetstar.com>

8. <http://www.vietjetair.com>

9. The specification of BPMN is managed by the Object Management Group <http://www.bpmn.org/>

10. Company AD has to wait for customer's confirmation sent via email. Latency caused by any delay in making such a confirmation by the customer could result in a final charge for rescheduling her airfare that is actually higher than the reschedule fees previously confirmed by the provider. For this reason, company AD puts a reasonable surcharge to cover this potential payment gap. The customer is entitled for a refund in case her actual reschedule fees finally go below the surcharge she paid.

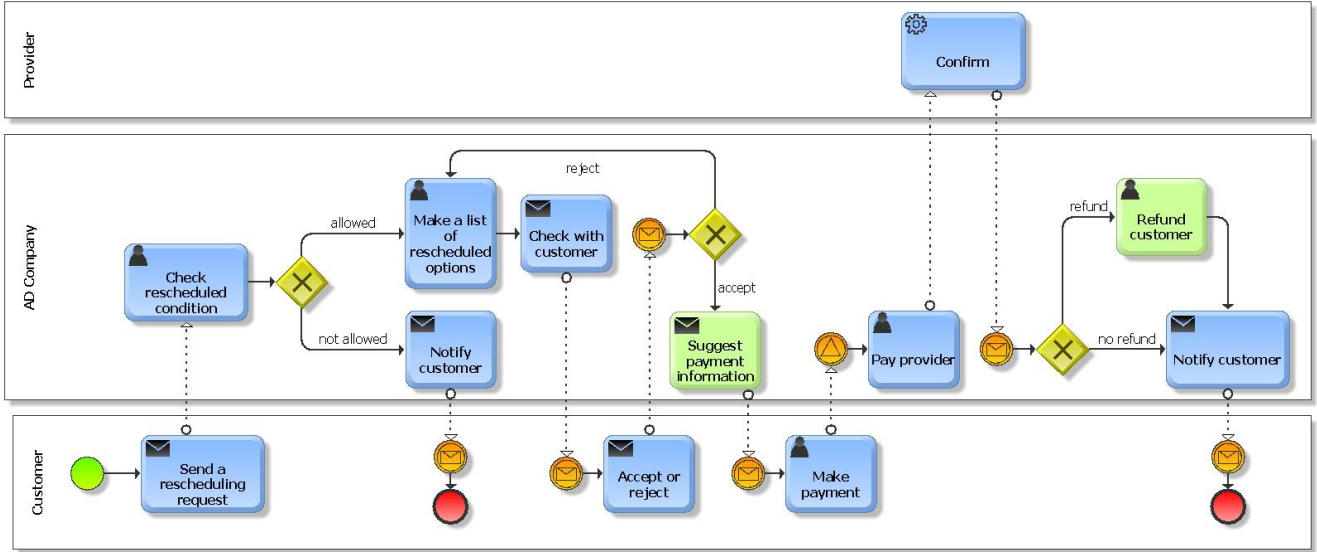


Figure 1. The as-is rescheduling airfare process in company AD

business process modeling from visualization forms (e.g., BPMN, UML) to a formal representation (ACP model).

3.2. Research Questions

In the context of business process modeling, BPMN is a notable language because it offers a standard and complete notation system and it is popular among domain experts [10]. However, it focuses too much on control flow to lack information of other aspects, such as addressing resource allocation, data flow, performance measurement. It leads to acquire extending BPMN for capturing the fields in question. For example, BPMN is expanded to capture resource perspective requirements [7] [8] and constraints [9]; to show the process execution [17]; to evaluate performance measures [5] [6]; to present data flow [10].

Among the above aspects, the artifact-centric approach is the most notable because data might be extracted as something meaningful that plays a driver role to improve the processes to be more effective. Currently, there has been some research in this field. They capture the flow of data (i.e., specifying the input/output objects of tasks) but lack of adequate information of artifacts (i.e., states and attributes of objects). It leads to the connection between processes and data is not enough tightness, consistency and preciseness.

That is the motivation to force us investigate how to augment the business process model (BPMN) with state machine and business artifacts. In this research direction, we have to solve the following problems: i) how to identify the impact of tasks on associated objects in business processes? (e.g., current states of the input/output objects) ii) how to determine the states of objects will be affected by changing of its attributes? iii) how to combine the such information to generate pre-conditions and post-conditions in ACP model (semi-)automatically?

4. Proposed Framework

4.1. Overview

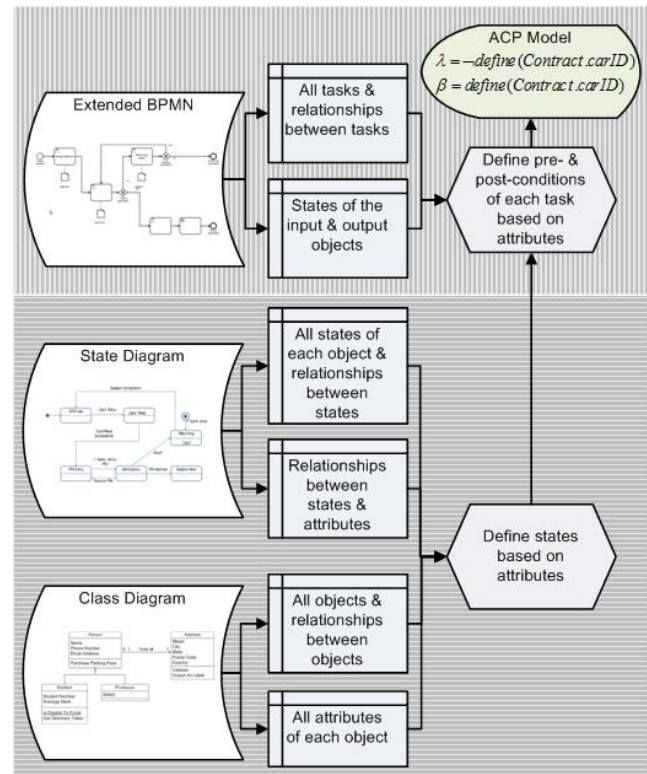


Figure 2. Overall framework

Figure 2 presents the overall picture of our approach in transformation of business process modeling to solve the research questions in Subsection 3.2. In our framework, we

collect the information from varied diagrams (e.g., state diagrams, BPMN diagrams) to enhance the accuracy and consistency of generated rules in ACP models (e.g., pre-conditions, post-conditions). The artifact classes provide list of all attributes of objects, as well as the states of them. One of the most important parts is state diagram that describes the relationships between states of an object, especially, each state provides the essential information of the associated attributes. All the aforementioned information are used to define the definition of states based on attribute expressions. In addition, business processes are presented in extended BPMN that shows the links between tasks, as well as identifies the current states of the input and output objects of each task. Ultimately, the pre- and post-conditions of ACP models could be generated in forms of state expressions or attribute expressions.

Our framework will transform the business process depicted in Figure 1 into what is presented in Table 2, 1, 4 & 3. In ACP model, each task is depicted as a 3-tuple of pre-condition, post-condition and service. The pre- and post-conditions could be defined as logical expressions of states or attributes that assures the consistency and accuracy of business process modeling. In the next steps, these expressions (pre- and post-conditions of tasks) will be combined with business rules (IF-THEN rules) and then compared with data mining rules (IF-THEN rules) in order for re-engineering business processes [3]. It is one of the most notable purposes of using ACP models in Industry 4.0. The rationale behind this transformation is that business processes are presented in the similar logical forms with other relevant aspects (e.g., artifacts, business rules, constraints) in order to make them understand each other (semi-)automatically for improving/enacting/measuring it more precisely and effectively.

4.2. Definitions

An artifact schema is defined as $M = \{C_1, C_2, \dots, C_n\}$ [12], where $C_i \in M (1 \leq i \leq n)$ is an artifact class. An artifact class C includes two ingredients: a set of attributes $A = \{a_1, a_2, \dots, a_x\}$, $a_i \in A (1 \leq i \leq x)$ and a set of states $S = \{s_1, s_2, \dots, s_y\} \cup \{s^{init}\}$, $s_i \in S (1 \leq i \leq y)$.

A business process P involves some scenario labels L , denoted by $P = \{L_1, L_2, \dots, L_m\}$. Each scenario label L_i contains a set of business tasks, presented as $L_i = \{t_1, t_2, \dots, t_\ell\}$.

We also adopt the concept of pre-condition and post-condition [12] in formally representing business tasks. From now on, we define a business task t as a 4-tuple of the following elements: *pre-condition* (λ), *post-condition* (β), *service* (ν) and *set of involved artifacts* (σ)

$$t = (\lambda, \beta, \nu, \sigma)$$

where λ and β are pre-condition and post-condition of the task in question, respectively; $\nu \in V$ is a service to be performed; σ is a set of artifact classes are involved in the task t : $\sigma = \{C_1, C_2, \dots, C_y\}$, where $C_i \in M (1 \leq i \leq y)$.

Definition 1 (Pre-condition & Post-condition). We argue that a pre-condition $t_i.\lambda$ and post-condition $t_i.\beta$ of a business task could be described formally in terms of attributes of the involved artifacts. We follow the rationale of design by contract [18] according to which state machine could be ignored, as opposed to what was originally proposed by the ACP [12].

We address that the pre-condition and post-condition are defined by higher-order logic formula supporting quantifier. For simplification, every variable expression in both conditions is linked to the others just by AND logical connection ($\wedge$). Each expression only contains attribute proposition which includes three types of forms as the following:

- the defined: $\text{defined}(C_j.a_k)$
- the undefined: $\neg \text{defined}(C_j.a_k)$
- the optional: $\text{optional}(C_j.a_k)$

where $i \in \{1, \dots, |L_h|\}$; $j \in \{1, \dots, |\sigma|\}$; $k \in \{1, \dots, |C_j.A|\}$;

$C_j \in \sigma$ is an artifact class; $C_j.a_k \in C_j.A$ is an attribute of class C_j ;

Example 1. In rescheduling process of company AD, we describe the pre-condition and post-condition of task "Check rescheduling condition" as the following:

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 $\lambda = \text{instate}(\text{CustomerSupportObject}, \text{initial})$ 
 $= \neg \text{defined}(\text{CustomerSupportObject}. \text{customerSupportID})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{oldBookingID})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{staffID})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{supportingType})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{startTime})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{finishTime})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{expectedReschedulingTime})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{newBookingID})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{customerDecision})$ 

 $\beta = \text{instate}(\text{CustomerSupportObject}, \text{opened})$ 
 $= \text{defined}(\text{CustomerSupportObject}. \text{customerSupportID})$ 
 $\wedge \text{defined}(\text{CustomerSupportObject}. \text{oldBookingID})$ 
 $\wedge \text{defined}(\text{CustomerSupportObject}. \text{staffID})$ 
 $\wedge \text{defined}(\text{CustomerSupportObject}. \text{supportingType})$ 
 $\wedge \text{defined}(\text{CustomerSupportObject}. \text{startTime})$ 
 $\wedge \text{defined}(\text{CustomerSupportObject}. \text{expectedReschedulingTime})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{newBookingID})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{customerDecision})$ 
 $\wedge \neg \text{defined}(\text{CustomerSupportObject}. \text{finishTime})$ 

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4.3. Implementation

We walk through the implementation of our framework. We illustrate it using the running example presented in Subsection 3.1. Let us begin by giving an extended BPMN diagram of rescheduling process as in Figure 3, the artifact classes of corresponding objects described in the third paragraph of Section 4.3, and state diagrams of associated objects in Figure 4, 5, 6.

The business process of rescheduling function is described in Figure 3 that is used for clarifying the process presented in Figure 1 by adding the input/output objects of tasks, as well as the current states of them. The information of the said objects are shown as the artifact classes in the

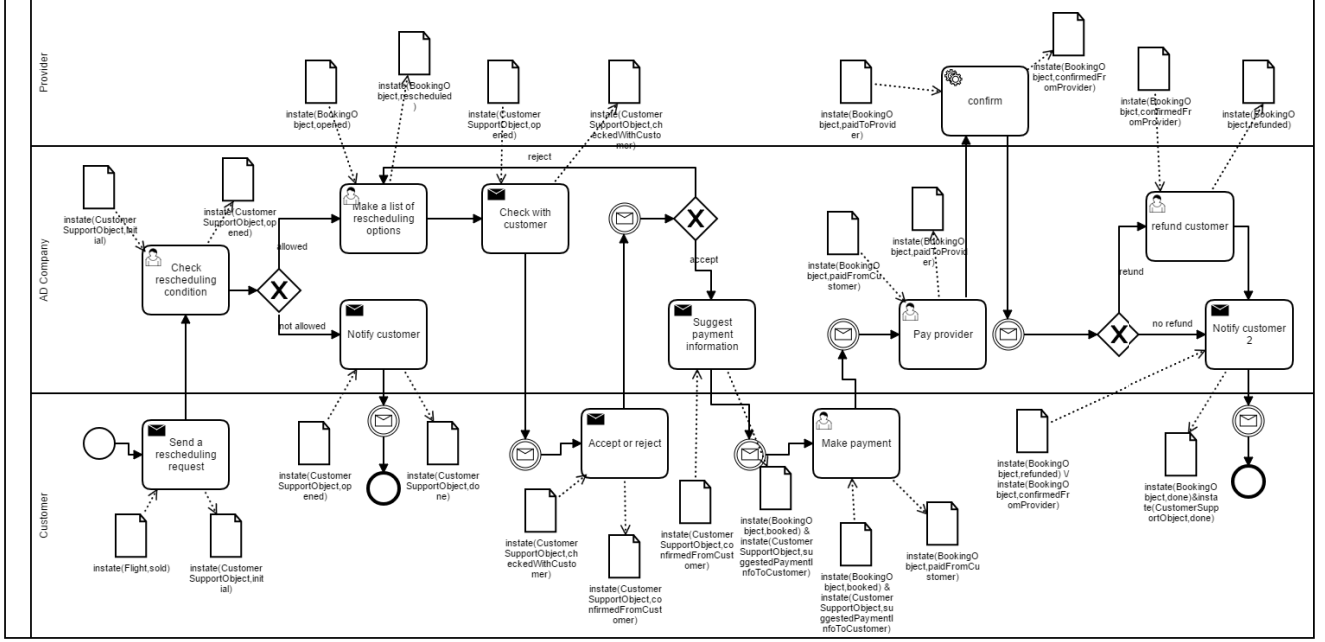


Figure 3. The extended BPMN of rescheduling process in company AD

TABLE 1. IN ARTIFACT *CustomerSupportObject*, ITS STATES COULD BE DEFINED BASED ON THE RELATIONSHIP WITH ITS ATTRIBUTES

$instate(CustomerSupportObject, initial)$	$= \neg defined(CustomerSupportObject.customerSupportID) \wedge \neg defined(CustomerSupportObject.oldBookingID) \wedge \neg defined(CustomerSupportObject.staffID) \wedge \neg defined(CustomerSupportObject.supportingType) \wedge \neg defined(CustomerSupportObject.startTime) \wedge \neg defined(CustomerSupportObject.finishTime) \wedge \neg defined(CustomerSupportObject.expectedReschedulingTime) \wedge \neg defined(CustomerSupportObject.newBookingID) \wedge \neg defined(CustomerSupportObject.customerDecision)$
$instate(CustomerSupportObject, opened)$	$= defined(CustomerSupportObject.customerSupportID) \wedge \neg defined(CustomerSupportObject.oldBookingID) \wedge defined(CustomerSupportObject.staffID) \wedge \neg defined(CustomerSupportObject.supportingType) \wedge \neg defined(CustomerSupportObject.startTime) \wedge defined(CustomerSupportObject.expectedReschedulingTime) \wedge \neg defined(CustomerSupportObject.newBookingID) \wedge \neg defined(CustomerSupportObject.customerDecision) \wedge \neg defined(CustomerSupportObject.finishTime)$
$instate(CustomerSupportObject, checkedWithCustomer)$	$= defined(CustomerSupportObject.customerSupportID) \wedge \neg defined(CustomerSupportObject.oldBookingID) \wedge defined(CustomerSupportObject.staffID) \wedge defined(CustomerSupportObject.supportingType) \wedge \neg defined(CustomerSupportObject.startTime) \wedge defined(CustomerSupportObject.expectedReschedulingTime) \wedge defined(CustomerSupportObject.newBookingID) \wedge \neg defined(CustomerSupportObject.customerDecision) \wedge \neg defined(CustomerSupportObject.finishTime)$
$instate(CustomerSupportObject, confirmedFromCustomer)$	$= defined(CustomerSupportObject.customerSupportID) \wedge \neg defined(CustomerSupportObject.oldBookingID) \wedge defined(CustomerSupportObject.staffID) \wedge defined(CustomerSupportObject.supportingType) \wedge \neg defined(CustomerSupportObject.startTime) \wedge defined(CustomerSupportObject.expectedReschedulingTime) \wedge defined(CustomerSupportObject.newBookingID) \wedge defined(CustomerSupportObject.customerDecision) \wedge \neg defined(CustomerSupportObject.finishTime)$
$instate(CustomerSupportObject, done)$	$= defined(CustomerSupportObject.customerSupportID) \wedge \neg defined(CustomerSupportObject.oldBookingID) \wedge defined(CustomerSupportObject.staffID) \wedge defined(CustomerSupportObject.supportingType) \wedge \neg defined(CustomerSupportObject.startTime) \wedge defined(CustomerSupportObject.finishTime) \wedge defined(CustomerSupportObject.expectedReschedulingTime) \wedge defined(CustomerSupportObject.newBookingID) \wedge defined(CustomerSupportObject.customerDecision)$

TABLE 2. IN ARTIFACT *Flight*, ITS STATES COULD BE DEFINED BASED ON THE RELATIONSHIP WITH ITS ATTRIBUTES

$instate(Flight, initial)$	$= \neg defined(Flight.flightID) \wedge \neg defined(Flight.departurePlace) \wedge \neg defined(Flight.arrivalPlace) \wedge \neg defined(Flight.departureTime) \wedge \neg defined(Flight.flightTime) \wedge \neg defined(Flight.flightDistance) \wedge \neg defined(Flight.airline) \wedge \neg defined(Flight.airplaneType) \wedge \neg defined(Flight.airplaneNumber) \wedge \neg defined(Flight.totalSeats) \wedge \neg defined(Flight.reservedSeats) \wedge \neg defined(Flight.vacantSeats) \wedge currentTime < Flight.departureTime$
$instate(Flight, opened)$	$= defined(Flight.flightID) \wedge \neg defined(Flight.departurePlace) \wedge \neg defined(Flight.arrivalPlace) \wedge \neg defined(Flight.departureTime) \wedge defined(Flight.flightTime) \wedge \neg defined(Flight.flightDistance) \wedge \neg defined(Flight.airline) \wedge \neg defined(Flight.airplaneType) \wedge defined(Flight.airplaneNumber) \wedge \neg defined(Flight.totalSeats) \wedge \neg defined(Flight.reservedSeats) \wedge \neg defined(Flight.vacantSeats) \wedge currentTime < Flight.departureTime \wedge Flight.reservedSeats < Flight.totalSeats$
$instate(Flight, soldout)$	$= defined(Flight.flightID) \wedge \neg defined(Flight.departurePlace) \wedge \neg defined(Flight.arrivalPlace) \wedge \neg defined(Flight.departureTime) \wedge defined(Flight.flightTime) \wedge \neg defined(Flight.flightDistance) \wedge \neg defined(Flight.airline) \wedge \neg defined(Flight.airplaneType) \wedge defined(Flight.airplaneNumber) \wedge \neg defined(Flight.totalSeats) \wedge \neg defined(Flight.reservedSeats) \wedge Flight.vacantSeats = 0 \wedge currentTime < Flight.departureTime \wedge Flight.reservedSeats = Flight.totalSeats$
$instate(Flight, closed)$	$= defined(Flight.flightID) \wedge \neg defined(Flight.departurePlace) \wedge \neg defined(Flight.arrivalPlace) \wedge \neg defined(Flight.departureTime) \wedge defined(Flight.flightTime) \wedge \neg defined(Flight.flightDistance) \wedge \neg defined(Flight.airline) \wedge \neg defined(Flight.airplaneType) \wedge defined(Flight.airplaneNumber) \wedge \neg defined(Flight.totalSeats) \wedge \neg defined(Flight.reservedSeats) \wedge \neg defined(Flight.vacantSeats) \wedge currentTime \geq Flight.departureTime$

third paragraph of Section 4.3. Additionally, the aforementioned states of objects are depicted in detail in Figure 4, 5, 6.

In the following, we address the primary artifact classes

that matter in the business process in question.

- $Flight = (\{flightID, departurePlace, arrivalPlace, departureTime, flightTime, flightDistance, airline, airplaneType, airplaneNumber, totalSeats,$

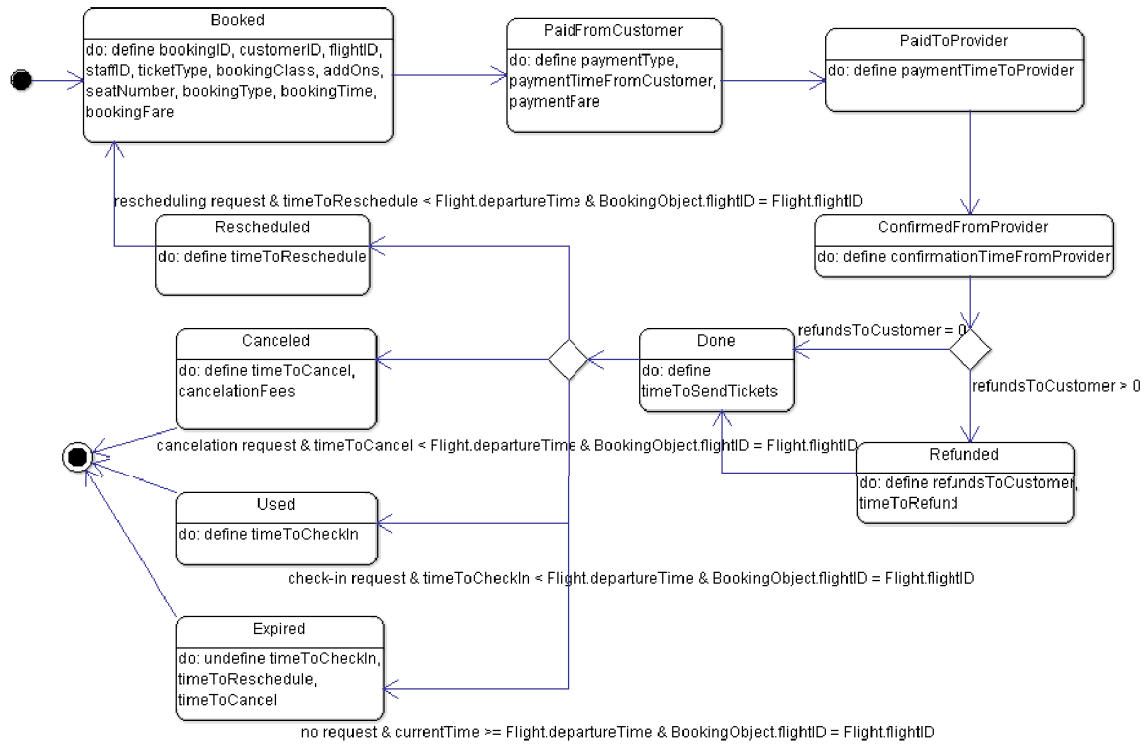


Figure 4. The state diagram of booking object in company AD

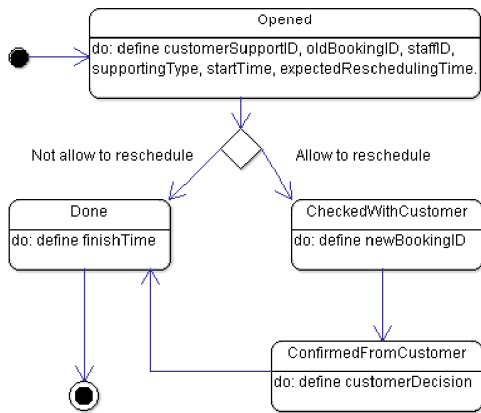


Figure 5. The state diagram of customer support object in company AD

reservedSeats, vacantSeats}, {initial, opened, soldout, closed}))

- Customer = ({customerID, age, gender, address, nationality}, ∅)
- Staff = ({staffID, age, gender, address, nationality, role, workingPeriod, performanceReview}, ∅)
- BookingObject = ({bookingID, customerID, flightID, staffID, ticketType, bookingClass, addOns, seatNumber, bookingType, bookingTime,

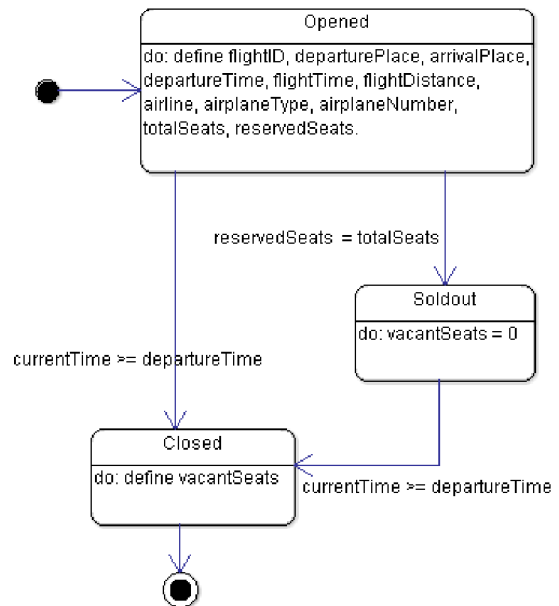


Figure 6. The state diagram of flight object in company AD

bookingFare, paymentType, paymentFare, paymentTimeFromCustomer, paymentTimeToProvider, confirmationTimeFromProvider, refundsToCustomer, timeToRefund, timeToCheckin, timeToCancel,

- timeToReschedule, cancelFees, timeToSendTickets},
{initial, opened, rescheduled, booked,
paidFromCustomer, paidToProvider,
confirmedFromProvider, refunded, done, used,
expired, rescheduled, canceled})
- CustomerSupportObject = ({customerSupportID,
oldBookingID, staffID, supportingType,
startTime, finishTime, expectedReschedulingTime,
newBookingID, customerDecision}, {initial, opened,
checkedWithCustomer, confirmedFromCustomer, done})

According to the artifact classes and state diagrams, we generate the definition of states of *Flight*, *CustomerSupportObject*, *BookingObject* as a form of attribute expressions in Tables 2, 1, 4, respectively. In addition, based on extended BPMN diagram represented in Figure 3, we produce the ACP model of rescheduling process described in Table 3. As a consequence, the pre- and post-conditions of ACP model are also presented as a form of state expressions based on the definition of state depicted in Tables 2, 1, 4.

We implemented our framework using the Orbital¹¹ library – a Java library supporting logic inference and Camunda BPM (Business Process Management) platform¹² library – an open-source platform for business process management. Our implementation¹³ has more than 2000 Java lines of code. It was done in the Eclipse¹⁴ platform that ran on a 64-bit Windows (Core i5, 6 GB of RAM). Taking the rescheduling process of company AD as the input, the system came out with the ACP model of rescheduling process as depicted in Tables 2, 1, 4, 3.

5. Related Work

In the literature, there has been a lot of research on extending BPMN for many different purposes. To capture advanced choreography scenarios, Decker and Puhmann [16] addressed the weaknesses of BPMN and proposed extensions to overcome these limitations and validated using the Services Interaction Patterns. In the resource perspective, BPMN is extended to support the modeling and visualization of resource perspective requirements [7] [8]. Moreover, they used the Object Constraint Language (OCL) embedded to BPMN for expressing resource allocation constraints [9].

To present the process execution, Momotko and Nowicki [17] defined a process instance (i.e. process execution) notation as an extension of BPMN to depict the process history, its current state and feasible future execution.

For measuring performance of business processes, an extension of BPMN meta-model [5] might show the performance analytic (e.g., time and cost) of business processes. It also provided new symbols to present divergent operational objects, data objects, resources, etc. By the same token,

11. Homepage of Orbital symbolaris.com/orbital

12. Homepage of Camunda <https://camunda.com>

13. Source code can be downloaded at <https://github.com/esp-lab/bpmn2acp>

14. Homepage of Eclipse eclipse.org

TABLE 3. RESCHEDULING PROCESS IN ARTIFACT-CENTRIC BUSINESS PROCESS MODELING LANGUAGE.

t₁ Send a rescheduling request	
Artifact classes	<i>Flight, CustomerSupportObject</i>
Pre-condition	<i>instate(Flight, opened)</i>
Service	<i>SendAReschedulingRequest(Flight, CustomerSupportObject)</i>
Post-condition	<i>instate(CustomerSupportObject, initial)</i>
t₂ Check rescheduling condition	
Artifact classes	<i>CustomerSupportObject, BookingObject</i>
Pre-condition	<i>instate(CustomerSupportObject, initial)</i>
Service	<i>CheckReschedulingCondition(CustomerSupportObject, BookingObject)</i>
Post-condition	<i>instate(CustomerSupportObject, opened)</i>
t₃ Notify customer	
Artifact classes	<i>CustomerSupportObject</i>
Pre-condition	<i>instate(CustomerSupportObject, opened)</i>
Service	<i>NotifyCustomer(CustomerSupportObject)</i>
Post-condition	<i>instate(CustomerSupportObject, done)</i>
t₄ Make a list of rescheduling options	
Artifact classes	<i>BookingObject, CustomerSupportObject</i>
Pre-condition	<i>instate(BookingObject, opened)</i>
Service	<i>MakeAListOfReschedulingOptions(BookingObject, CustomerSupportObject)</i>
Post-condition	<i>instate(BookingObject, rescheduled)</i>
t₅ Check with customer	
Artifact classes	<i>CustomerSupportObject, BookingObject</i>
Pre-condition	<i>instate(CustomerSupportObject, opened)</i>
Service	<i>CheckWithCustomer(CustomerSupportObject, BookingObject)</i>
Post-condition	<i>instate(CustomerSupportObject, checkedWithCustomer)</i>
t₆ Accept or reject	
Artifact classes	<i>CustomerSupportObject, BookingObject</i>
Pre-condition	<i>instate(CustomerSupportObject, checkedWithCustomer)</i>
Service	<i>AcceptOrReject(CustomerSupportObject, BookingObject)</i>
Post-condition	<i>instate(CustomerSupportObject, confirmedFromCustomer) ∧ instate(BookingObject, booked)</i>
t₇ Suggest payment information	
Artifact classes	<i>CustomerSupportObject, BookingObject</i>
Pre-condition	<i>instate(CustomerSupportObject, confirmedFromCustomer)</i>
Service	<i>SuggestPaymentInformation(CustomerSupportObject, BookingObject)</i>
Post-condition	<i>instate(CustomerSupportObject, done)</i>
t₈ Make payment	
Artifact classes	<i>BookingObject, CustomerSupportObject</i>
Pre-condition	<i>instate(BookingObject, booked) ∧ instate(CustomerSupportObject, done)</i>
Service	<i>MakePayment(BookingObject, CustomerSupportObject)</i>
Post-condition	<i>instate(BookingObject, paidFromCustomer)</i>
t₉ Pay provider	
Artifact classes	<i>BookingObject</i>
Pre-condition	<i>instate(BookingObject, paidFromCustomer)</i>
Service	<i>PayProvider(BookingObject)</i>
Post-condition	<i>instate(BookingObject, paidToProvider)</i>
t₁₀ Confirm	
Artifact classes	<i>BookingObject</i>
Pre-condition	<i>instate(BookingObject, paidToProvider)</i>
Service	<i>Confirm(BookingObject)</i>
Post-condition	<i>instate(BookingObject, confirmedFromProvider)</i>
t₁₁ Refund customer	
Artifact classes	<i>BookingObject</i>
Pre-condition	<i>instate(BookingObject, confirmedFromProvider)</i>
Service	<i>RefundCustomer(BookingObject)</i>
Post-condition	<i>instate(BookingObject, refunded)</i>
t₁₂ Notify customer 2	
Artifact classes	<i>BookingObject, CustomerSupportObject</i>
Pre-condition	<i>(instate(BookingObject, refunded) ∨ instate(BookingObject, confirmedFromProvider) ∧ BookingObject.refundsToCustomer = 0)</i>
Service	<i>NotifyCustomer(BookingObject, CustomerSupportObject)</i>
Post-condition	<i>instate(BookingObject, done)</i>

Korherr and List [6] proposed an extension for the Event-Driven Process Chain (EPC) and BPMN with process goals and performance measures (e.g., quality, cost, cycle time).

To describe data flow in business processes, Lohmann and Nyolt [10] investigated to extend BPMN for expressing data aspects in business processes.

6. Conclusion

The main contribution of this research is the valuable knowledge regarding augmenting the business processes

TABLE 4. IN ARTIFACT *BookingObject*, ITS STATES COULD BE DEFINED BASED ON THE RELATIONSHIP WITH ITS ATTRIBUTES

[illegible]

with state machine and business artifacts through solving three research questions presented in Section 3.2. We proposed a formal framework for generating ACP models based on extended BPMN, state diagrams and artifact classes that will support to improve, or enact, or measure business

processes. Our framework was applied to company AD for transferring its rescheduling process from a visualization form (BPMN) to a formal representation (ACP model) and using it as the input for the next step in business process re-engineering.

Despite its benefits, our approach has some limitations, and there is possibly room for improvement. The main limitation is that the proposed framework is applied to solve the problem in narrow domain of business process re-engineering. However, in the future, the aforementioned framework could be expanded to more various modeling languages for applying to many different areas.

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